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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 05/15/2026 |
| **PI Name:** | Martin, Bryn Ph.D. | | |
| **Project Title:** | Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study) | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | bmartin@neelatx.com | **Fax:** | FY |
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| **City:** | Moscow | **State:** | ID |
| **Zip Code:** | 83843 | **Congressional District:** | 1 |
| **Comments:** | PI moved to Alcyone Therapeutics, Inc., in late 2020 from University of Idaho (Ed., December 2021) PI moved to Neela Therapeutics in late 2025 from Alcyone (Ed, November 2025) | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C |
| **Start Date:** | 12/08/2021 | **End Date:** | 12/07/2025 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/07/2025 per NSSC information (Ed., 11/14/24).<br>NOTE: End date of new grant 80NSSC22K0023 is 12/07/2024 per grant information documents (Ed., December 2021).<br>NOTE: End date of previous grant number 80NSSC20K0920 changed to 10/12/2020 per JSC Grants Admin/NSSC information, due to PI move and new grant established (Ed., 12/10/21). | | |
| **Key Personnel Changes/Previous PI:** | There have not been any key personnel changes in the last reporting period. | | |
| **COI Name (Institution):** | Kramer, Larry M.D. ( University of Texas Health Science Center at Houston )<br>Laurie, Steven Ph.D. ( KBR/NASA Johnson Space Center )<br>Macias, Brandon Ph.D. ( NASA Johnson Space Center )<br>Marshall-Goebel, Karina Ph.D. ( KBR/NASA Johnson Space Center )<br>Williams, Michael M.D. ( University of Washington, Seattle )<br>Loerch, Linda Ph.D. ( NASA Johnson Space Center )<br>Hasan, Khader ( University of Texas Health Sciences Center ) | | |
| **Grant/Contract No.:** | 80NSSC22K0023 | | |

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| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | NOTE: Continuation of "Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)," grant 80NSSC20K0920, with the same Principal Investigator (PI) Dr. Bryn Martin.<br>The proposed research plan builds on our group's ongoing research to quantify ophthalmic and intracranial changes in long duration spaceflight astronauts using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. We propose to use identical techniques currently implemented pre- and post-spaceflight to also quantify how long-duration bed rest and countermeasures applied during bed rest impact ophthalmic and intracranial structures. It is critically essential to use identical techniques that we have already applied in astronauts to determine if structural alterations seen in bed rest are similar to what our team has already quantified in astronauts with spaceflight associated neuro-ocular syndrome (SANS). Similarly, it is essential to use the identical techniques to quantify the capability of a countermeasure to mitigate structural alterations in bed rest. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The principal scientific impact of the OPTICS study is the demonstration that 30-day head-down tilt bed rest reproduces several key SANS-relevant MRI signatures with quantitative fidelity, particularly anterior optic globe shift, upward brain shift, and intracranial CSF redistribution. This validates 30-day HDT as a useful (though not perfect) ground-based analog for evaluating future countermeasures, and establishes a small set of well-characterized imaging endpoints that future countermeasure studies can target. The negative result for thigh cuffs and LBNP tested in this protocol provides important guidance that single-modality, intermittent countermeasures during HDT may be insufficient and that combined or continuous interventions warrant evaluation.<br>Beyond spaceflight, the brain-shift and CSF-shift biomarkers developed under this award have direct translational application to terrestrial CSF disorders, particularly idiopathic normal pressure hydrocephalus (iNPH) and related CSF disorders. |
| **Task Progress:** | Long-duration spaceflight produces structural and physiological changes in the eyes and brain of astronauts, collectively termed Spaceflight Associated Neuro-ocular Syndrome (SANS). SANS is an important medical risk for future missions to the Moon and Mars, and a major NASA Human Research Program priority is to identify both the underlying mechanisms and effective countermeasures. The OPTICS study addressed three SANS gap areas: (SANS1) etiological mechanisms and risk factors for in-flight and post-flight ocular changes; (SANS12) whether ground-based analogs can simulate SANS; and (SANS13) identification of preventative and treatment countermeasures.<br>OPTICS combined two key strengths into a single analysis pipeline. First, 48 healthy volunteers underwent a 30-day, 6-degree head-down tilt (HDT) bedrest protocol, a well-established Earth-bound analog for the headward fluid shift astronauts experience in microgravity, with subgroups receiving thigh-cuff or lower-body negative pressure (LBNP) countermeasures. Second, the same MRI imaging and quantitative analysis pipelines were applied to 26 astronauts who had completed long-duration spaceflight missions, with imaging at multiple time points before, during (HDT only), and after the experimental period. This parallel design allowed direct comparison of magnitude and recovery time course between the analog and the actual spaceflight experience.<br><br>During the period of performance, the OPTICS team developed three novel quantitative MRI methods that filled measurement gaps. First, an automated technique was developed to measure forward shift of the eyeball within its orbit using whole-head T1-weighted MRI and bone segmentation-based registration. Second, a method was developed to quantify upward (and three-dimensional) shift of the brain's center of mass inside the skull. Third, a semi-automated pipeline was developed for measuring spinal cord and cerebrospinal fluid (CSF) geometry along the entire spine. All three methods were validated against ground-truth manual segmentations and applied to all study cohorts.<br><br>Five key findings emerged. (1) Anterior optic globe shift was significant in both 30-day HDT participants and astronauts at return + 3 days. This represents the first quantitative evidence that low-grade forward shift of the eye may be part of the SANS picture, and that 30-day HDT reproduces this signature. (2) Upward brain shift was significant in both cohorts but larger and longer-lasting in astronauts; recovery to baseline took up to 180 days post-flight, while HDT participants showed partial recovery by 12 days post-bedrest. (3) Ventricular CSF volume increased in both cohorts, supporting a consistent fluid-redistribution signature across the analog and spaceflight. (4) Spinal CSF cross-sectional area decreased during 30-day HDT and returned to baseline 12 days post-HDT. (5) The thigh-cuff and LBNP countermeasures tested at the HDT-29 timepoint did not produce a measurable effect on any of the parameters studied.<br><br>Findings were disseminated via a peer-reviewed publication in npj Microgravity (Warthen et al., 2025) and a 2025 publication in J Appl Physiol on normal variation of brain volumetrics, CSF dynamics, and ocular structures in healthy participants over two years. A peer-reviewed manuscript on anterior globe shift in astronauts was preprinted in 2026 and is currently under review. A spinal CSF manuscript is in late-stage preparation. The team also presented progress at the 2024, 2025, and 2026 NASA HRP Investigators' Workshops, the 16th Meeting of the Hydrocephalus Society (Nagoya, Japan, 2024), and the Hydrocephalus World Congress (Toulouse, France, 2025). [Ed. Note: For complete citations, see the Cumulative Bibliography.]<br><br>The OPTICS study provides quantitative comparison between an Earth-based microgravity analog and long-duration spaceflight, using identical imaging analysis pipelines across cohorts. The work identifies anterior globe shift, upward brain shift, and intracranial CSF redistribution as the SANS-relevant MRI biomarkers most reliably reproduced by 30-day HDT, and potentially useful endpoints for evaluating future countermeasures. The negative result for thigh cuff and LBNP is itself informative, suggesting that single-modality intermittent countermeasures during a 30-day analog may not be sufficient to reverse the headward fluid-shift signature, and that more aggressive or continuous countermeasures may be required for Mars-class missions. The MRI-based brain-shift and CSF-shift biomarkers also have potential applications beyond spaceflight, including idiopathic normal pressure hydrocephalus, other CSF disorders, and therapeutic interventions. |

| Bibliography Type: | Description: (Last Updated: 06/30/2026) |
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| Articles in Other Journals or Periodicals | Martin B, Warthen K, Sater S, Roberts C, Reilly M, Katz S, Kramer L, Hasan K, Carter K, Williams M, Macias B, Laurie S. "Anterior ocular globe shift observed in astronauts after long-duration spaceflight." Research Square preprint server. Posted March, 15, 2026. https://doi.org/10.21203/rs.3.rs-8912940/v1 , Mar-2026 |
| Articles in Peer-reviewed Journals | Warthen KG, Sater SH, Kramer LA, Hasan KM, Williams MA, Young M, Macias BR, Laurie SS, Martin BA "Brain and cerebrospinal fluid 3D center of mass shift after spaceflight." npj Microgravity. 2025 May 8;11(1):14. https://doi.org/10.1038/s41526-025-00468-x ; PubMed PMID: 40335486; PubMed Central PMCID: PMC12058997 , May-2025 |
| Articles in Peer-reviewed Journals | Carter KJ, Smith JD, Huang AS, Stern C, Kramer LA, Hasan KM, Chen X, Martin BA, Young M, Brunstetter TJ, Macias BR, Laurie SS. "Repeat exposures to spaceflight or bedrest and spaceflight-associated neuro-ocular syndrome findings." JAMA Ophthalmol. 2026 Jun 18. Online ahead of print. http://dx.doi.org/10.1001/jamaophthalmol.2026.2140 ; PMID: 42313423; PMCID: PMC13280762 , Jun-2026 |